

PROJECT REPORT:

STUDENT MANAGEMENT SYSTEM

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Aim:

To develop a **Student Management System** using C++ that helps manage student records efficiently by performing operations like adding, displaying, searching, updating, and deleting student information.

Problem Statement:

Managing student records manually is time-consuming and prone to errors. Educational institutions need a simple digital system to store and manage student details such as roll number, name, marks, and grades.

This project solves the problem by creating a menu-driven application that stores student data systematically and performs operations quickly

Key Features :

- . Add new student record
- . Display all student records
- . Search student by roll number
- . Update student details
- . Delete student record
- . Calculate percentage
- . Assign grades
- . File handling for data storage

Scope of Project:

This project can be used in:

- . Schools
- . Colleges
- . Coaching institutes
- . Training centers

Algorithm :

Step 1:

Start the program

Step 2:

Declare arrays for storing:

- Roll numbers
- Names
- Marks

Step 3:

Open file `students.txt`

Step 4:

Load previously saved student records

Step 5:

Display menu

1. Add Student
2. Display Students
3. Search Student
4. Save to File
5. Exit

Step 6:

- Take user choice

- **If choice = 1 (Add Student)**

- Input roll number
- Input name
- Input marks
- Store in arrays
- Increase count

- **If choice = 2 (Display Students)**

- Traverse arrays
- Display details
- Calculate grade

- **If choice = 3 (Search Student)**

- Input roll number
- Search in array
- Display record if found

- **If choice = 4 (Save File)**

- Open file
- Write all student records

- **If choice = 5 (Exit)**

- Save data
 - End program

Code:

```
#include<iostream>
#include<fstream>
using namespace std;

int main() {
    int rollNo[100];
    string name[100];
    float marks[100];
    int count = 0;
    int choice, searchRoll;

    // Load data from file
    ifstream infile("students.txt");
    while(infile >> rollNo[count] >> name[count] >> marks[count]) {
        count++;
    }
    infile.close();

    while(true) {
        cout << "\n\n==== STUDENT MANAGEMENT SYSTEM =====";
        cout << "\n1. Add Student";
        cout << "\n2. Display Students";
        cout << "\n3. Search Student";
        cout << "\n4. Save Records";
        cout << "\n5. Exit";
        cout << "\nEnter Choice: ";
        cin >> choice;

        switch(choice) {

            case 1:
                cout << "\nEnter Roll Number: ";
                cin >> rollNo[count];

                cout << "Enter Name: ";
                cin >> name[count];

                cout << "Enter Marks: ";
                cin >> marks[count];

                count++;
            }
        }
    }
```

```

        cout << "Student Added Successfully!";
        break;

case 2:

    if(count == 0) {
        cout << "\nNo Records Found!";
    }
    else {
        for(int i = 0; i < count; i++) {
            cout << "\n\nRoll Number: " << rollNo[i];
            cout << "\nName: " << name[i];
            cout << "\nMarks: " << marks[i];
            cout << "\nPercentage: " << marks[i] << "%";

            if(marks[i] >= 90)
                cout << "\nGrade: A";
            else if(marks[i] >= 75)
                cout << "\nGrade: B";
            else if(marks[i] >= 50)
                cout << "\nGrade: C";
            else
                cout << "\nGrade: Fail";

            cout << "\n-----";
        }
    }
    break;

case 3:

    cout << "\nEnter Roll Number to Search: ";
    cin >> searchRoll;

    bool found;
    found = false;

    for(int i = 0; i < count; i++) {
        if(rollNo[i] == searchRoll) {
            cout << "\nStudent Found!";
            cout << "\nRoll Number: " << rollNo[i];
            cout << "\nName: " << name[i];
            cout << "\nMarks: " << marks[i];
            found = true;
        }
    }
}

```

```

        if(found == false)
            cout << "\nStudent Not Found!";
        break;

    case 4: {
        ofstream outfile("students.txt");

        for(int i = 0; i < count; i++) {
            outfile << rollNo[i] << " "
                << name[i] << " "
                << marks[i] << endl;
        }

        outfile.close();
        cout << "\nRecords Saved Successfully!";
        break;
    }
}

```

```

    case 5: {
        ofstream outfile("students.txt");

        for(int i = 0; i < count; i++) {
            outfile << rollNo[i] << " "
                << name[i] << " "
                << marks[i] << endl;
        }

        outfile.close();

        cout << "\nData Saved. Exiting Program...";
        return 0;
    }

    default:
        cout << "\nInvalid Choice!";
}

return 0;
}

```


Sample output:

===== STUDENT MANAGEMENT SYSTEM =====

1. Add Student
2. Display Students
3. Search Student
4. Save Records
5. Exit

Enter Choice: 1

Enter Roll Number: 101

Enter Name: Poonam

Enter Marks: 91

Student Added Successfully!

Roll Number: 101

Name: Poonam

Marks: 91

Percentage: 91%

Grade: A

Enter Choice: 4

Records Saved Successfully!

Enter Choice: 5

Data Saved. Exiting Program...

Conclusion:

The Student Management System successfully manages student information digitally. It reduces manual work, improves accuracy, and allows quick access to student records.

This project demonstrates practical implementation of:

- . Object-Oriented Programming
- . Classes and Objects
- . Functions
- . File handling
- . Conditional statements
- . Loops